Quicksort exercises

07-E01 Lomuto partition trace

Difficulty: Foundational.

Partition A=[7,3,8,3,6,2] with final element 2 as pivot. List every exchange and the final pivot index.

07-E02 A 1:9 recursion bound

Difficulty: Advanced.

Each partition costs n and the largest child has size at most 9n/10. Prove total O(n log n) by a recursion tree.

07-E03 Comparing four-rank endpoints

Difficulty: Advanced.

For four ordered distinct elements, analyze whether random Quicksort compares elements 1 and 4. Verify indicator probability 1/2.

07-E04 Equal keys and random pivots

Difficulty: Research.

All n keys equal 5. Lomuto sends keys $\leq$pivot left; each pivot is chosen randomly and moved to the end. Derive exact comparisons, identify the missing premise in "expected O(n log n) on every input," and give a duplicate-key repair.

07-E05 Smaller-side recursion stack

Difficulty: Research.

Recurse only on the smaller partition and continue the larger in a while loop. Prove worst-case O(log n) stack. Use an always-maximum pivot to show why time can remain $\Theta(n^2)$.

07-E06 Negative and repeated keys

Difficulty: Advanced.

Use 1-based Lomuto on A=[−1,4,0,4,2] with final pivot 2. List i,j and the array at each step, q, and the three final regions. Does partition imply sorted sides?

Quicksort exercises (continued)

07-E07 Outside pivots, comparison events, and the cost model

Difficulty: Research. The input contains distinct keys 1 through 7 in any initial order. Choose a uniform pivot in every current subproblem, using this chapter's Lomuto contract.
(1) Find the probability that keys 2 and 5 ever compare directly. List all four possible first pivots within {2,3,4,5}; explain why the denominator is not 7.
(2) Find the conditional probability if the first pivot of the whole execution is 7. What if it is 3? Explain how the condition changes the event.
(3) A three-way implementation uses successive < and > tests, so classifying a non-pivot key may require two Boolean comparisons. Can $2(n+1)H_n - 4n$ still be used without qualification as the exact number of Boolean comparisons? Also distinguish the recursive boundaries here and in the Data Structures sorting chapter.

Work independently before opening the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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